On the constant in the inequality $\mathcal{J} \le C\,\|\cdot\|_K$

(Janson's Lemma 1)

1 Setting and results

Let α be an additive commutative monoid and A_0, A_1 quasinorms on α with values in $[0,\infty]$ (a `Couple`), written $\|\cdot\|_0$, $\|\cdot\|_1$. For $t>0$, $\nu\in\mathbb{Z}$ and $x,u\in\alpha$ put

$$K(t,x)=\inf_{x=x_0+x_1}\bigl(\|x_0\|_0+t\|x_1\|_1\bigr),\qquad K_\nu(x)=K(2^\nu,x),\qquad J_\nu(u)=\max\bigl(\|u\|_0,\ 2^\nu\|u\|_1\bigr)$$

(`kNorm` and `jNorm`), and $\Delta=\{x\in\alpha:\|x\|_0<\infty,\ \|x\|_1<\infty\}$ (`A.inf.finiteLocus`). Fix $0<\theta<1$ and $0<q\le\infty$. For $f\colon\mathbb{Z}\to[0,\infty]$, $\|f\|_{\ell^q}$ is the ℓ^q-norm (`eLpNorm f q Measure.count`). We compare

$$\begin{aligned}
\|x\|_K &= \bigl\|(2^{-\nu\theta}K_\nu(x))_{\nu\in\mathbb{Z}}\bigr\|_{\ell^q} && (\texttt{discreteKMethod}),\\
\|x\|_{K,c} &= \Bigl(\int_0^\infty \bigl(t^{-\theta}K(t,x)\bigr)^q\,\frac{dt}{t}\Bigr)^{1/q} && (\texttt{kMethod}),\\
\mathcal{J}(x) &= \inf_u \bigl\|(2^{-\nu\theta}J_\nu(u_\nu))_{\nu\in\mathbb{Z}}\bigr\|_{\ell^q} && (\texttt{jInfNorm}).
\end{aligned}$$

For $q=\infty$, $\|x\|_{K,c}$ is the essential supremum of $t^{-\theta}K(t,x)$ over $t>0$. In $\mathcal{J}(x)$ the infimum runs over all finitely supported $u\colon\mathbb{Z}\to\alpha$ with $\sum_\nu u_\nu=x$. We consider two classes of couples.

(S) α carries a modulus $x\mapsto|x|$ with the refined Riesz decomposition (R) of the companion note `jInfNorm_le_discreteK.tex` (`Modulus.IsDecomposable`), and A_0, A_1 are solid (`IsSolid`): $|x|\le|y|$ implies $\|x\|_i\le\|y\|_i$.

(N) α is an abelian group, and A_0, A_1 are norms in the sense of groups: $\|x+y\|_i\le\|x\|_i+\|y\|_i$ and $\|-x\|_i=\|x\|_i$.

Class (N) is the normed case of Janson's setting [1], where the spaces are quasi-normed abelian groups. The upper bound for class (S) is Theorem 1 of the companion note:

Theorem 1. *Let the couple be of class (S) and $0<q\le\infty$. Then $\mathcal{J}(x)\le 2^{\min(\theta,1-\theta)}\|x\|_K$ for every $x\in\Delta$.*

Corollary 3 shows that this constant is optimal for $q=\infty$, and that no constant $C\le 1$ is possible, also for $\|\cdot\|_{K,c}$ and large finite q. The example behind it is a couple on $\mathbb{R}$ whose quasinorms only detect whether an element is zero. Then every nonzero piece of a J-decomposition costs as much as x itself, so splitting x over several levels does not help, while $K(t,x)$ still takes the minimum of the two norms at each level. Proposition 4 gives the constant $1+2^{\min(\theta,1-\theta)}$ for class (N) and $q\ge 1$, by differences of near-optimal decompositions.

2 An example

For a statement P let $\mathbf{1}(P)$ be 1 if P holds and 0 otherwise.

Proposition 2. *Let $0 < \theta < 1$. On $\alpha = \mathbb{R}$, with the modulus $|\cdot|$ (`Real.instModulus`), put*

$$\|v\|_0 = 2^\theta\, \mathbf{1}(v \neq 0), \qquad \|v\|_1 = \mathbf{1}(v \neq 0).$$

(a) This couple belongs to both classes (S) and (N), and $1 \in \Delta$.

(b) $K(t,1) = \min(2^\theta, t)$ for every $t > 0$.

(c) $\mathcal{J}(1) = 2^{\min(\theta,1-\theta)}$ for every $q \in (0,\infty]$.

(d) $\|1\|_K = 1$ if $q = \infty$, and $\|1\|_K^q = S_q := \big(1 - 2^{-(1-\theta)q}\big)^{-1} + \big(1 - 2^{-\theta q}\big)^{-1}$ if $q < \infty$.

(e) $\|1\|_{K,c} = 2^{\theta(1-\theta)}$ if $q = \infty$, and $\|1\|_{K,c} = 2^{\theta(1-\theta)}\big(\theta(1-\theta)q\big)^{-1/q}$ if $q < \infty$.

Proof. (a) Each $\|\cdot\|_i$ has the form $c_i\, \mathbf{1}(v \neq 0)$ with $c_i > 0$. It vanishes exactly at 0 and is even. If $v + w \neq 0$, then $v \neq 0$ or $w \neq 0$, so $\|v+w\|_i \leq \|v\|_i + \|w\|_i$. Hence the couple is of class (N), and each $\|\cdot\|_i$ is an `EQuasinorm` with constant 1. For solidity, let $|v| \leq |w|$. If $w = 0$, then $v = 0$; in both cases $\|v\|_i \leq \|w\|_i$. The modulus $|\cdot|$ on $\mathbb{R}$ satisfies (R) (`Real.instIsDecomposable`), so the couple is of class (S). Both norms are finite, so $1 \in \Delta$.

(b) In a decomposition $1 = x_0 + x_1$ at least one part is nonzero, and the cost is $2^\theta \mathbf{1}(x_0 \neq 0) + t\, \mathbf{1}(x_1 \neq 0)$. The decompositions $1 = 1+0$ and $1 = 0+1$ cost 2^θ and t, and a decomposition with both parts nonzero costs $2^\theta + t$. Hence $K(t,1) = \min(2^\theta, t)$.

(d) Put $a_\nu = 2^{-\nu\theta} K_\nu(1) = 2^{-\nu\theta} \min(2^\theta, 2^\nu)$. If $\nu \leq 0$, then $2^\nu \leq 1 < 2^\theta$, so $a_\nu = 2^{\nu(1-\theta)}$. If $\nu \geq 1$, then $2^\nu \geq 2 > 2^\theta$, so $a_\nu = 2^{(1-\nu)\theta}$. Both are geometric sequences with largest value $a_0 = a_1 = 1$, which gives the case $q = \infty$. For $q < \infty$,

$$\sum_{\nu\in\mathbb{Z}} a_\nu^q = \sum_{k\geq 0} 2^{-k(1-\theta)q} + \sum_{k\geq 0} 2^{-k\theta q} = S_q.$$

(e) By (b), the function $t \mapsto t^{-\theta} \min(2^\theta, t)$ equals $t^{1-\theta}$ on $(0, 2^\theta]$ and $2^\theta t^{-\theta}$ on $[2^\theta, \infty)$. It is continuous, increasing on the first interval and decreasing on the second, so its (essential) supremum is its value $2^{\theta(1-\theta)}$ at $t = 2^\theta$. For $q < \infty$,

$$\int_0^{2^\theta} t^{(1-\theta)q}\, \frac{dt}{t} + 2^{\theta q} \int_{2^\theta}^\infty t^{-\theta q}\, \frac{dt}{t} = \frac{2^{\theta(1-\theta)q}}{(1-\theta)q} + \frac{2^{\theta q}\, 2^{-\theta^2 q}}{\theta q} = \frac{2^{\theta(1-\theta)q}}{\theta(1-\theta)q}.$$

(c) If $u \neq 0$, then $J_\nu(u) = \max(2^\theta, 2^\nu)$, so $2^{-\nu\theta} J_\nu(u) = c_\nu$, where $c_\nu = 2^{(1-\nu)\theta}$ for $\nu \leq 0$ and $c_\nu = 2^{\nu(1-\theta)}$ for $\nu \geq 1$. The sequence c decreases on $\nu \leq 0$ and increases on $\nu \geq 1$, so $\min_\nu c_\nu = \min(c_0, c_1) = \min(2^\theta, 2^{1-\theta}) = 2^{\min(\theta,1-\theta)}$.

Let u be admissible for $\mathcal{J}(1)$. Since $\sum_\nu u_\nu = 1 \neq 0$, some u_μ is nonzero. The ℓ^q-norm of a sequence is at least each of its entries, so the cost of u is at least $c_\mu \geq 2^{\min(\theta,1-\theta)}$. Conversely, a sequence with a single nonzero entry has that entry as its ℓ^q-norm. Placing 1 at level 0 or at level 1 therefore costs $c_0 = 2^\theta$ or $c_1 = 2^{1-\theta}$. □

Corollary 3. *Let $0 < \theta < 1$, and suppose that $\mathcal{J}(x) \leq C\|x\|_K$ for all couples of class (S), or for all couples of class (N), and all $x \in \Delta$.*

(i) If $q = \infty$, then $C \geq 2^{\min(\theta,1-\theta)}$. Hence the constant of Theorem 1 is optimal for $q = \infty$. In particular, the constant 2^θ, which the proof of Theorem 1 gives when it uses only the shift $\nu \mapsto \nu + 1$, is optimal for $\theta \leq 1/2$.

(ii) If $q < \infty$, then $C \geq 2^{\min(\theta,1-\theta)} S_q^{-1/q}$. This bound tends to $2^{\min(\theta,1-\theta)}$ as $q \to \infty$, so $C > 1$ for all large q. For $\theta = 1/2$ the bound equals $\big((2^{q/2}-1)/2\big)^{1/q}$, which exceeds 1 if and only if $q > 2\log_2 3 \approx 3.17$.

If instead $\mathcal{J}(x) \leq C\|x\|_{K,c}$, then

$$C \geq 2^{\min(\theta,1-\theta)^2}\big(\theta(1-\theta)q\big)^{1/q} \qquad (\text{the second factor read as } 1 \text{ if } q = \infty),$$

and this bound exceeds 1 whenever $q \geq 1/(\theta(1-\theta))$, including $q = \infty$.

Proof. Apply the inequality to the couple of Proposition 2 and $x = 1$, and divide (c) by (d) or (e). For (ii), $S_q \to 2$ as $q \to \infty$, so $S_q^{1/q} \to 1$. For $\theta = 1/2$, $S_q = 2/(1 - 2^{-q/2})$, so $2^{q/2}/S_q = (2^{q/2} - 1)/2$, which exceeds 1 if and only if $2^{q/2} > 3$. For the continuous method, $\min(\theta, 1-\theta) - \theta(1-\theta) = \min(\theta, 1-\theta)^2$: if $\theta \leq 1/2$, the difference is $\theta - \theta + \theta^2$, and otherwise it is $(1-\theta) - \theta(1-\theta) = (1-\theta)^2$. Since $2^{\min(\theta,1-\theta)^2} > 1$, the bound exceeds 1 as soon as $\theta(1-\theta)q \geq 1$. □

3 The argument by differences

Janson's proof of his Lemma 1 truncates infinite J-decompositions and does not track constants. The following argument starts from near-optimal K-decompositions, as in Theorem 1, but takes differences of consecutive decompositions instead of using (R).

Proposition 4. *For couples of class (N), every $q \in [1,\infty]$ and every $x \in \Delta$, $\mathcal{J}(x) \leq \big(1 + 2^{\min(\theta,1-\theta)}\big)\|x\|_K$.*

Proof. We prove the bound with $1 + 2^\theta$; the bound with $1 + 2^{1-\theta}$ is Remark 5. Put $a_\mu = 2^{-\mu\theta}K_\mu(x)$. The decompositions $x = x + 0$ and $x = 0 + x$ show $K_\mu(x) \leq \min(\|x\|_0, 2^\mu\|x\|_1)$, which is finite because $x \in \Delta$.

Fix $N \geq 1$ and $\delta > 0$, and put $W_N = \{-N, \dots, N-1\}$. For every $\mu \in \mathbb{Z}$ choose $x = y_\mu + z_\mu$ with

$$\|y_\mu\|_0 + 2^\mu\|z_\mu\|_1 \leq \varepsilon_\mu := K_\mu(x) + \delta. \tag{1}$$

Since $y_\mu + z_\mu = x = y_{\mu+1} + z_{\mu+1}$, we have $y_{\mu+1} - y_\mu = z_\mu - z_{\mu+1}$. Define $v\colon \mathbb{Z} \to \alpha$ by

$$v_{-N} = y_{-N+1}, \qquad v_\nu = y_{\nu+1} - y_\nu \quad (-N < \nu < N-1), \qquad v_{N-1} = z_{N-1},$$

and $v_\nu = 0$ for $\nu \notin W_N$. Since $N \geq 1$, the indices $-N$ and $N-1$ are distinct; for $N = 1$ the middle range is empty. The middle terms telescope, so

$$\sum_{\nu\in\mathbb{Z}} v_\nu = y_{-N+1} + (y_{N-1} - y_{-N+1}) + z_{N-1} = y_{N-1} + z_{N-1} = x,$$

and v is admissible in the definition of $\mathcal{J}$. Put $E_{-N} = 2^{-N}\|x\|_1$, $E_{N-1} = \|x\|_0$ and $E_\nu = 0$ otherwise. We claim that for every $\nu \in W_N$

$$J_\nu(v_\nu) \leq \varepsilon_\nu + \varepsilon_{\nu+1} + E_\nu. \tag{2}$$

It suffices to bound both entries of $J_\nu(v_\nu)$ by the right-hand side. We use the triangle inequality, symmetry, and (1) in the forms $\|y_\mu\|_0 \leq \varepsilon_\mu$ and $2^\mu\|z_\mu\|_1 \leq \varepsilon_\mu$.

- If $-N < \nu < N-1$, then $\|v_\nu\|_0 \le \|y_{\nu+1}\|_0 + \|y_\nu\|_0 \le \varepsilon_{\nu+1} + \varepsilon_\nu$ and $2^\nu\|v_\nu\|_1 \le 2^\nu\|z_\nu\|_1 + \frac{1}{2}\,2^{\nu+1}\|z_{\nu+1}\|_1 \le \varepsilon_\nu + \frac{1}{2}\varepsilon_{\nu+1}$.
- If $\nu = -N$, then $\|v_\nu\|_0 = \|y_{-N+1}\|_0 \le \varepsilon_{-N+1}$. Since $y_{-N+1} = x - z_{-N+1}$, $2^{-N}\|y_{-N+1}\|_1 \le 2^{-N}\|x\|_1 + \frac{1}{2}\,2^{-N+1}\|z_{-N+1}\|_1 \le E_{-N} + \frac{1}{2}\varepsilon_{-N+1}$.
- If $\nu = N-1$, then, since $z_{N-1} = x - y_{N-1}$, $\|v_\nu\|_0 \le \|x\|_0 + \|y_{N-1}\|_0 \le E_{N-1} + \varepsilon_{N-1}$, and $2^{N-1}\|z_{N-1}\|_1 \le \varepsilon_{N-1}$.

Multiply (2) by $2^{-\nu\theta} = 2^\theta\,2^{-(\nu+1)\theta}$. With $w_\nu = 2^{-\nu\theta}\mathbf{1}(\nu \in W_N)$ and $e_\nu = 2^{-\nu\theta}E_\nu$, this gives

$$b_\nu := 2^{-\nu\theta}J_\nu(v_\nu) \le a_\nu + 2^\theta a_{\nu+1} + 2\delta\, w_\nu + e_\nu$$

for $\nu \in W_N$, and also for $\nu \notin W_N$, where $b_\nu = 0$. Since $q \ge 1$, Minkowski's inequality and the shift invariance of $\|\cdot\|_{\ell^q}$ give

$$\mathcal{J}(x) \le \|b\|_{\ell^q} \le (1 + 2^\theta)\|x\|_K + 2\delta\|w\|_{\ell^q} + e_{-N} + e_{N-1}.$$

The sequence w is finitely supported and does not depend on δ, so $\|w\|_{\ell^q} < \infty$, and letting $\delta \to 0$ removes the middle term. Finally $e_{-N} = 2^{N\theta}2^{-N}\|x\|_1 = 2^{-(1-\theta)N}\|x\|_1$ and $e_{N-1} = 2^{-(N-1)\theta}\|x\|_0$ tend to 0 as $N \to \infty$, because $0 < \theta < 1$ and $x \in \Delta$. □

Remark 5 (The constant $1 + 2^{1-\theta}$). Take instead $v_{-N} = y_{-N}$, $v_\nu = y_\nu - y_{\nu-1} = z_{\nu-1} - z_\nu$ for $-N < \nu < N-1$, and $v_{N-1} = z_{N-2}$. These sum to $y_{N-2} + z_{N-2} = x$. Since $2^\nu\|z_{\nu-1}\|_1 = 2 \cdot 2^{\nu-1}\|z_{\nu-1}\|_1 \le 2\varepsilon_{\nu-1}$, the same case analysis gives $J_\nu(v_\nu) \le \varepsilon_\nu + 2\varepsilon_{\nu-1} + E_\nu$. Because $2 \cdot 2^{-\nu\theta} = 2^{1-\theta}\,2^{-(\nu-1)\theta}$, this becomes $b_\nu \le a_\nu + 2^{1-\theta}a_{\nu-1} + 3\delta\, w_\nu + e_\nu$, and the rest of the proof is unchanged.

Remark 6 (p-normed couples). Suppose that $\|x+y\|_i^p \le \|x\|_i^p + \|y\|_i^p$ for some $0 < p \le 1$, and $\|-x\|_i = \|x\|_i$. Taking p-th powers in the proof, as Janson indicates, gives $b_\nu^p \le a_\nu^p + 2^{\theta p}a_{\nu+1}^p$ up to terms that vanish in the limit. Put $s = \min(p, q)$. If $q \ge p$, Minkowski's inequality in $\ell^{q/p}$ applies to b^p; if $q < p$, the map $t \mapsto t^{q/p}$ is subadditive, which gives $b_\nu^q \le a_\nu^q + 2^{\theta q}a_{\nu+1}^q$. In both cases $\mathcal{J}(x) \le (1 + 2^{\theta s})^{1/s}\|x\|_K$, and likewise with $1-\theta$. This constant grows like $2^{1/s}$ as $s \to 0$. For a general quasinorm, the passage to an equivalent p-norm (Aoki–Rolewicz) costs a further factor that depends on the quasi-triangle constant.

Corollary 7. *For couples of class (N) and $q = \infty$, the best constant C in $\mathcal{J}(x) \le C\|x\|_K$ satisfies $2^{\min(\theta,1-\theta)} \le C \le 1 + 2^{\min(\theta,1-\theta)}$.*

Proof. Corollary 3(i) and Proposition 4. □

4 Remarks

Remark 8 (Where the summand 1 comes from). In Proposition 4 each piece is a difference of two near-optimal decompositions, so the triangle inequality bounds its norms by costs at two levels, $\varepsilon_\nu + \varepsilon_{\nu+1}$. After weighting this is $a_\nu + 2^\theta a_{\nu+1}$, and Minkowski's inequality adds the two contributions. In Theorem 1, (R) produces pieces with $|u_\nu| \le |y_{\nu+1}|$ and $|u_\nu| \le |z_\nu|$, so each of the two norms of u_ν is bounded through a single decomposition, and $K_\nu \le K_{\nu+1}$ leaves only $2^\theta a_{\nu+1}$. The triangle inequalities of A_0, A_1 and ℓ^q are not used there, which is why that constant does not depend on q or on quasi-triangle constants.

Remark 9 (Normed vector spaces). The quasinorms of Proposition 2 are not homogeneous. For homogeneous norms, spreading x over several levels lowers $\mathcal{J}(x)$. For example, let $\|v\|_0 = c|v|$ and $\|v\|_1 = |v|$ on $\mathbb{R}$, and $q = \infty$. Then $K(t,1) = \min(c,t)$ as before. Put $w_\nu = 2^{-\nu\theta}\max(c, 2^\nu)$. Every admissible u satisfies

$$1 \le \sum_\nu |u_\nu| \le \sup_\nu (w_\nu |u_\nu|) \sum_\nu w_\nu^{-1},$$

and the truncations of $(w_\nu^{-1})_\nu$ to $|\nu| \le M$, normalized to sum 1, approach equality as $M \to \infty$, so $\mathcal{J}(1) = \bigl(\sum_\nu w_\nu^{-1}\bigr)^{-1}$. For $c = \sqrt{2}$ and $\theta = 1/2$ this gives $\mathcal{J}(1) = (\sqrt{2}-1)/2 \approx 0.207$, while $\|1\|_K = 1$.

Numerically, a grid search over c and θ in these one-dimensional couples finds no larger ratio $\mathcal{J}(1)/\|1\|_K$, for $q = \infty$ and $q = 1$. A search over pairs of polyhedral norms on $\mathbb{R}^2$ and $\mathbb{R}^3$, with both sides computed by linear programming on truncated level ranges, found ratios of at most about 0.21. We do not know whether $\mathcal{J} \le C\|\cdot\|_K$ holds with some $C < 1$ for all couples of normed vector spaces.

References

[1] S. Janson, *On interpolation of multi-linear operators*, in: M. Cwikel et al. (eds.), Function Spaces and Applications, Lecture Notes in Math. 1302, Springer, Berlin, 1988, pp. 290–302.